

UjenziXform Trader Invoicing System (TIS) — User & Operator Manual

Document type: User / operator manual (certification submission) **Applicant:** UjenziXform Solution
Product: UjenziXform Trader Invoicing System (TIS) v1.0.0 **Platform:** UjenziXform (construction materials marketplace, Kenya) **Reference:** UJX-TIS-KRA-ARCH-2026-001 **Document version:** 1.0 **Date:** 3 June 2026

Status: Applicant / in development — KRA sandbox access not yet assigned.

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1. Purpose and audience

This manual describes how **operators and end users** work with the **UjenziXform Trader Invoicing System (TIS)** — the eTIMS compliance layer embedded in UjenziXform purchase order workflows.

Audience	What this manual covers
Suppliers	Enter KRA PIN and legal business details; submit orders to eTIMS; view fiscal receipts
Builders (buyers)	Maintain billing identity; receive and verify fiscal receipts
Administrators	Onboard vendors, run OSCU initialization, monitor submissions, retry failures
KRA reviewers	Understand day-to-day operator workflows during certification

TIS is **not** a standalone accounting product. It maps **purchase orders (POs)** to KRA **SalesReq** JSON and submits them via **POST /invoices** once KRA assigns sandbox or production access.

2. Roles and access

Role	TIS capabilities
Supplier	Edit own KRA settings; manually submit PO to eTIMS; view fiscal receipt on accepted orders
Builder	View fiscal receipt on own POs; optional KRA PIN in profile for B2B invoices
Admin / super_admin	Full TIS Integrator Hub; vendor onboarding; submission ops; OSCU init; API console
Guest / unauthenticated	No TIS API access

All outbound KRA calls require a signed-in user with role `supplier`, `admin`, or `super_admin`. KRA credentials are **never** visible in the browser.

3. Before you start

3.1 Prerequisites (per supplier taxpayer)

Before the first fiscal invoice can be submitted:

Requirement	Where configured
Valid KRA PIN	Supplier → KRA / eTIMS settings
Legal business name (matches KRA registration)	Same panel
Branch code (<code>bhfId</code> , e.g. <code>00</code>)	Same panel — from KRA sandbox or production
Device serial (<code>dvcSr1No</code>)	Same panel — from KRA after OSCU initialization
Item codes on every PO line	Supplier catalog / materials (<code>etims_item_code</code>)
PO status accepted or confirmed	Order workflow (not quote-pending)

3.2 Current platform status

Item	Status
TIS application software	Ready for sandbox testing
KRA eTIMS sandbox credentials	Pending — subject of integrator application
Target sandbox URL	<code>https://etims-api-sbx.kra.go.ke</code>
Integrator certification	Not held

Until KRA assigns sandbox access, connection tests may return **“eTIMS proxy not configured”**. Configuration is performed by administrators per the Installation & Deployment Guide.

4. Supplier: KRA tax identity setup

Navigation: Supplier dashboard → **Invoices & documents** → **KRA / eTIMS settings** tab **Component:** `SupplierEtimsSettingsPanel`

4.1 Fields to complete

Field	Purpose
Legal business name	Printed on fiscal receipt; maps to <code>traderName</code>
KRA PIN	Supplier taxpayer ID; maps to <code>traderTin</code> / <code>sellerPin</code>
VAT registration status	Internal compliance context
Physical business address	Invoice and audit reference
Invoice contact phone / email	Operational contact
Branch code	KRA branch office ID (<code>bhfId</code>)
Business place code	KRA business place reference where applicable
Device serial	OSCU device serial after initialization
Default payment type	Maps to KRA payment type code (default <code>01</code>)
Invoice notes	Optional footer text on supplier-facing documents

4.2 Save and test connection

1. Enter all required fields and click **Save**.
2. Click **Test connection** (when sandbox is configured) to verify `etims-proxy` reachability.
3. Resolve any KRA PIN validation errors before accepting new orders.

PIN format: Validated by `isValidKraPin()` — must conform to KRA PIN rules before checkout and submission.

5. Builder: buyer profile for fiscal invoices

Navigation: Profile → **Edit profile**

Field	Purpose
KRA PIN	Maps to <code>customerPin</code> on B2B invoices
Billing / company name	Maps to <code>customerName</code>

If the builder has no KRA PIN, TIS may still submit with buyer name only, depending on KRA sandbox rules for B2C sales. B2B transactions should always include a valid buyer PIN.

6. Purchase order → fiscal invoice flow

TIS submits `POST /invoices` from a **purchase order**, not from a separate internal invoice module.

6.1 Automatic submission paths

User action	Result
Cart → Buy Now	PO created with status <code>confirmed</code> → automatic TIS submit
Materials grid → Buy Now	Same as above

6.2 Manual submission paths

User action	Result
Supplier Order Management	Supplier clicks Submit to KRA eTIMS on an accepted PO
Admin TIS / eTIMS tools	Staff retry or test submit on a specific PO

6.3 Quote workflow

POs in quote-pending statuses (`quote_created` , `quote_received_by_supplier` , `quoted` , etc.) **block** automatic submission. After the builder accepts the quote and the PO reaches an accepted/confirmed status, the supplier or admin may submit manually.

6.4 Gates (submission blocked if any fail)

Gate	Error stored on
Missing or invalid supplier KRA PIN / legal name	<code>purchase_orders.etims_error</code>
PO still in draft, pending, or quote status	Same
PO line missing <code>etims_item_code</code>	Same
eTIMS proxy not configured	Same

6.5 Successful submission outcome

On success, the platform stores on the PO:

- `etims_validated_at` — timestamp of KRA acceptance
- `etims_response` — JSON snapshot (verification URL, SCU reference)
- Entry in `tis_submission_log` — audit trail

The buyer and supplier can then open the **fiscal receipt view**.

7. Viewing the fiscal receipt and QR code

Component: `EtimsFiscalReceiptView`

After a successful KRA response, the fiscal receipt displays:

Element	Description
Issuer name and KRA PIN	Supplier taxpayer
Trader invoice number	PO number (<code>traderInvoiceNo</code>)
SCU / control unit fields	From KRA response
Line items and totals	From PO
Verification URL	KRA public validation link
QR code	Encodes verification URL for mobile scanning

Operator action: Confirm the QR code scans correctly and opens the KRA verification page during sandbox certification.

8. Credit notes

Credit notes reverse or adjust a prior sale.

Requirement	Detail
Prior sale	PO must have a successful eTIMS sale (<code>etims_validated_at</code> set)
Receipt type	<code>receiptTypeCode: R</code>
Reference	<code>traderOrgInvoiceNo</code> links to original sale invoice number
Endpoint	Same <code>POST /invoices</code> via <code>etims-proxy</code>

Supplier action: Use the credit note action on the supplier invoice hub for the original PO (when enabled after sandbox assignment).

9. Administrator: TIS Integrator Hub

Navigation: Admin dashboard → **TIS Integrator** tab **Component:** `TisIntegratorHub`

The hub has four sub-tabs:

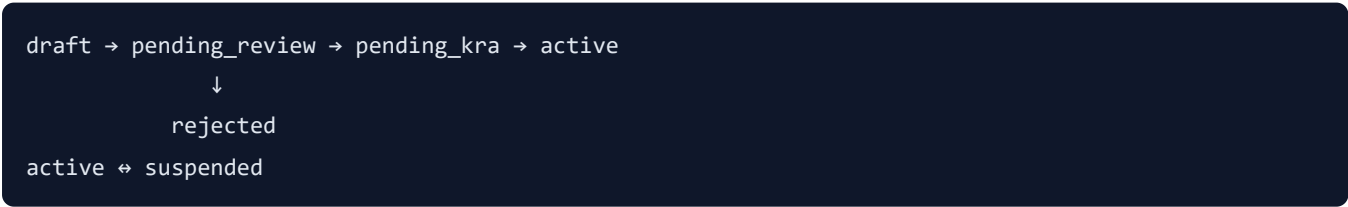
Tab	Purpose
Platform & certification	Integrator metadata, environment status, KRA certification checklist
Vendor onboarding	Per-supplier onboarding lifecycle
Submission ops	Monitor and retry failed PO submissions
Integrator API	Allowlisted REST calls to KRA sandbox (items, customers, reference data)

This hub is separate from the **eTIMS test** tab, which is used for low-level proxy smoke tests.

10. Administrator: vendor onboarding

Table: `tis_vendor_onboarding` **Panel:** `TisVendorOnboardingPanel`

10.1 Lifecycle states



Status	Meaning
draft	Supplier profile incomplete
pending_review	Awaiting admin review of tax identity and catalog
pending_kra	Awaiting OSCU init, item registration, or KRA steps
active	Cleared for sandbox/production invoicing
suspended	Temporarily blocked
rejected	Not approved for TIS

10.2 Admin procedure

1. Open **Vendor onboarding** in the TIS Integrator Hub.
2. Review supplier KRA PIN, legal name, branch, and device serial.
3. Advance status to `pending_kra` when internal review passes.
4. Complete OSCU initialization (§12) and item registration.
5. Set status to `active` before the supplier's first live sandbox invoice.

11. Administrator: submission operations and retry

Panel: `TisSubmissionOpsPanel` **Audit table:** `tis_submission_log`

11.1 Monitoring

Data source	What to check
<code>tis_submission_log</code>	Per-submission status, error message, response snapshot
<code>purchase_orders.etims_error</code>	Latest failure on a specific order
<code>purchase_orders.etims_validated_at</code>	Confirms successful fiscalisation

11.2 Retry procedure

1. Open **Submission ops** or the admin PO submit card (`EtimsPurchaseOrderSubmitCard`).
2. Identify POs with `etims_error` populated and no `etims_validated_at` .
3. Fix the root cause (missing item code, invalid PIN, wrong PO status).
4. Click **Submit to KRA eTIMS** or **Retry**.
5. Confirm a new `tis_submission_log` entry with status `success` .

12. Administrator: OSCU initialization

Library: `tisOscuInitialization.ts` **KRA endpoint:** `POST selectInitOsdInfo` (via `etims-proxy`)

12.1 Required inputs

Field	Description
<code>tin</code>	Vendor/supplier KRA PIN
<code>bhfId</code>	Branch office code (e.g. <code>00</code>)
<code>dvcSr1No</code>	KRA-assigned device serial for sandbox

12.2 Procedure

1. Ensure Edge secrets point to KRA sandbox (`ETIMS_BASE_URL=https://etims-api-sbx.kra.go.ke`).
2. Open **Integrator API** or the OSCU initialization form in the admin hub.
3. Enter supplier PIN, branch, and device serial.
4. Submit — KRA returns `cmcKey` , `dvcId` , and branch name.
5. Store device ID on the supplier record; **never** store communication keys in plain text in the database.

13. Administrator: Integrator API console

Panel: `TisIntegratorApiConsole`

Used during sandbox certification for:

Operation	Typical path
Reference data sync	GET endpoints (countries, tax codes, item classes)
Item registration	<code>POST /items</code>
Customer registration	<code>POST /customers</code>
Manual invoice test	<code>POST /invoices</code>

All paths must appear on the allowlist in `supabase/functions/_shared/etimsPathAllowlist.ts` . Requests exceeding 400 KB are rejected.

14. Common errors and resolution

Symptom	Likely cause	Resolution
"eTIMS proxy not configured"	Edge secrets missing	Admin: set <code>ETIMS_BASE_URL</code> , <code>ETIMS_BASIC_USER</code> , <code>ETIMS_BASIC_PASSWORD</code>
"Invalid KRA PIN"	PIN format or mismatch	Supplier: correct PIN in KRA settings
"Missing item code on line ..."	Catalog not mapped	Admin/supplier: set <code>etims_item_code</code> on product
"PO status blocks submission"	Quote not accepted	Complete quote workflow first
HTTP 403 from proxy	User lacks role	Assign <code>supplier</code> , <code>admin</code> , or <code>super_admin</code>
KRA rejection on <code>/invoices</code>	Payload or master data	Check SalesReq mapping; register items/customers on sandbox

15. Support and escalation

Level	Contact
End-user (supplier/builder)	UjenziXform platform support — <code>_[Complete before submission]_</code>
TIS / integrator technical	UjenziXform Solution — Baragetuny Plaza Left Wing 3rd Floor Room 10, Ronald Ngala Street, Eldoret. P. O. Box 4146 - 30100 Eldoret
KRA eTIMS	eTIMS portal and integrator certification unit

For certification testing, administrators should retain screenshots of successful submissions, fiscal receipts with QR codes, and `tis_submission_log` entries as evidence.

16. Document control

Version	Date	Changes
1.0	2026-06-03	Initial release for KRA optical disk submission

Related documents:

- `TIS_ETIMS_INTEGRATION_ARCHITECTURE.md` — full technical architecture
- `INSTALLATION_DEPLOYMENT_GUIDE.md` — deployment and Edge configuration
- `CERTIFICATION_TEST_PLAN.md` — sandbox test cases